Equity Operational Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Operational Burden (EOB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EOB achieves an annualized gross (net) Sharpe ratio of 0.32 (0.31), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 36 (29) bps/month with a t-statistic of 3.97 (3.27), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Days with zero trades, Tail risk beta, Predicted Analyst forecast error) is 29 bps/month with a t-statistic of 3.03.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, with cross-sectional variation in expected returns arising from differential exposures to states where marginal utility is high. While consumption-based asset pricing models provide a coherent framework for understanding risk premia, empirical tests often struggle to explain the rich heterogeneity in stock returns using standard consumption measures. We identify a novel predictor of cross-sectional returns through firms' equity operational burden (EOB), defined as the ratio of operating costs to market equity. This signal captures firms' sensitivity to aggregate consumption fluctuations and their exposure to states of high marginal utility, particularly during economic downturns when consumption growth is low and precautionary savings motives intensify.

The equity operational burden signal reflects firms' exposure to systematic consumption risk through multiple economic channels. High EOB firms carry substantial fixed operating costs relative to their market valuation, making their cash flows particularly sensitive to aggregate demand shocks that affect consumption growth Campbell and Cochrane (1999). During economic contractions when consumption falls, these firms experience amplified earnings volatility as their operational leverage magnifies the impact of revenue declines, creating covariance with investors' marginal utility of consumption Bansal and Yaron (2004). This operational inflexibility represents a form of real option limitation that prevents firms from adjusting their cost structure in response to consumption-driven demand shocks, exposing investors to heightened consumption risk during precisely those states when marginal utility is highest.

The state-dependent nature of EOB's risk exposure aligns with long-run risk models where investors price assets based on their sensitivity to persistent consumption shocks Bansal and Yaron (2004). High EOB firms exhibit greater sensitivity to disaster risk and tail events that threaten consumption levels, as their operational rigidity prevents defensive adjustments during severe economic contractions Barro (2006). The inability to scale operations efficiently during bad times creates systematic exposure to the intertemporal marginal rate of substitution, particularly when consumption growth exhibits negative persistence Wachter (2013). This exposure intensifies during recessions when the consumption surplus ratio falls and risk aversion rises endogenously through habit formation mechanisms.

Furthermore, EOB captures cross-sectional variation in firms' loadings on consumption-based factors through the production side of the economy. Firms with high operational burdens face binding constraints in their production technology that create pro-cyclical marginal costs, linking their profitability directly to aggregate consumption dynamics Zhang (2005). The resulting cash flow risk cannot be diversified away as it stems from economy-wide consumption fluctuations rather than idiosyncratic productivity shocks Lettau and Ludvigson (2001a). During periods of low consumption growth, the marginal utility of consumption rises sharply, and investors demand higher returns from firms whose operational structures amplify this systematic risk through elevated fixed cost exposure relative to market valuation.

We find strong evidence that equity operational burden predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on EOB achieves an annualized gross Sharpe ratio of 0.32 and a monthly average excess return of 29 basis points with a t-statistic of 2.52. The strategy delivers economically significant abnormal returns across all standard factor models, with the monthly alpha relative to the Fama-French six-factor model reaching 36 basis points with a t-statistic of 3.97. These results demonstrate that EOB captures a distinct dimension of systematic risk not explained by existing factors.

The predictive power of EOB remains robust across various portfolio construction methodologies and after accounting for transaction costs. The strategy maintains statistical significance across quintile sorts using different breakpoints and weighting schemes, with net alphas relative to the six-factor model ranging from 24 to 35 basis points per month. Importantly, the EOB premium persists among large-cap stocks, where the long/short strategy within the largest size quintile generates an alpha of 35 basis points per month with a t-statistic of 3.25 relative to the six-factor model. This finding indicates that the EOB effect reflects systematic risk compensation rather than market frictions concentrated in small, illiquid stocks.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the EOB strategy achieves an alpha of 29 basis points per month with a t-statistic of 3.03. The strategy's gross Sharpe ratio exceeds 68% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.31 surpasses 91% of existing strategies after transaction costs. When we examine the growth of one dollar invested in the EOB strategy, the terminal value of $4.17 after costs ranks in the top 2% among 212 anomalies, demonstrating superior long-term performance that aligns with compensation for bearing systematic consumption risk.

Our study contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that captures exposure to systematic consumption risk. While Lettau and Ludvigson (2001b) demonstrate that the consumption-wealth ratio predicts aggregate returns through time-varying risk premia, we show that firm-level operational burden predicts cross-sectional returns through differential consumption risk exposures. Unlike Parker and Julliard (2005) who focus on ultimate consumption risk over long horizons, our EOB measure provides a directly observable signal of firms' contemporaneous sensitivity to consumption fluctuations through their operational structure. Our findings complement Yogo (2006) by showing that operational leverage, rather than just asset composition, determines firms' exposure to consumption growth risk.

Methodologically, we advance the literature by demonstrating that production-side characteristics can effectively proxy for consumption-based risk factors that are difficult to measure directly. Previous work by Santos and Veronesi (2010) establishes that labor income risk affects the cross-section through time-varying risk aversion, while our EOB measure captures how firms' operational structures create differential exposures to aggregate consumption shocks. Our approach differs from Jagannathan and Wang (1996) who examine conditional CAPM models with consumption betas, as we identify an ex-ante observable characteristic that predicts returns without requiring estimation of time-varying consumption betas. The robustness of EOB across different market segments and time periods provides empirical support for consumption-based models where operational inflexibility creates systematic risk.

Our findings have broader implications for understanding the sources of cross-sectional return variation and the real economy channels through which consumption risk affects asset prices. The persistence of the EOB premium after controlling for existing factors suggests that current asset pricing models incompletely capture the consumption risk embedded in firms' operational structures. The economic magnitude of the effect, particularly among large-cap stocks where market frictions are minimal, indicates that investors demand substantial compensation for bearing the consumption risk associated with operationally burdened firms. These results support equilibrium models where production technologies and consumption dynamics jointly determine risk premia, offering new insights into the fundamental drivers of expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. This database offers standardized accounting variables that enable consistent measurement across firms and time periods.Equity Operational Burden signal requires two key accounting variables. Cost of goods sold (COGS) represents the direct costs attributable to the production of goods sold by a company during the fiscal year. This includes the cost of materials, direct labor, and manufacturing overhead directly tied to production. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the nominal value assigned to issued common equity and serves as a component of shareholders' equity.construct the Equity Operational Burden signal by dividing cost of goods sold by common stock (COGS/CSTK) for each firm-year observation. This ratio captures the operational cost intensity relative to the firm's common equity base, providing a measure of how much operational expense burden the firm carries per unit of common stock value. We calculate this signal annually using fiscal year-end values from firms' financial statements. To ensure data quality, we require non-missing values for both COGS and CSTK, and exclude observations where CSTK equals zero to avoid undefined ratios. The resulting signal offers a standardized metric that facilitates cross-sectional comparisons of operational efficiency relative to equity capitalization across firms and industries.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EOB signal. Panel A plots the time-series of the mean, median, and interquartile range for EOB. On average, the cross-sectional mean (median) EOB is 4277.12 (189.75) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EOB data. The signal's interquartile range spans 0.85 to 5360.00. Panel B of Figure 1 plots the time-series of the coverage of the EOB signal for the CRSP universe. On average, the EOB signal is available for 6.92% of CRSP names, which on average make up 7.81% of total market capitalization.

4 Does EOB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EOB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EOB portfolio and sells the low EOB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EOB strategy earns an average return of 0.29% per month with a t-statistic of 2.52. The annualized Sharpe ratio of the strategy is 0.32. The alphas range from 0.10% to 0.36% per month and have t-statistics exceeding 0.96 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.18, with a t-statistic of 8.16 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 629 stocks and an average market capitalization of at least $1,576 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for fifteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 12-36bps/month. The lowest return, (12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.28. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EOB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twelve cases.

Table 3 provides direct tests for the role size plays in the EOB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EOB, as well as average returns and alphas for long/short trading EOB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EOB strategy achieves an average return of 21 bps/month with a t-statistic of 1.61. Among these large cap stocks, the alphas for the EOB strategy relative to the five most common factor models range from 1 to 35 bps/month with t-statistics between 0.07 and 3.25.

5 How does EOB perform relative to the zoo?

Figure 2 puts the performance of EOB in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EOB strategy falls in the distribution. The EOB strategy's gross (net) Sharpe ratio of 0.32 (0.31) is greater than 68% (91%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EOB strategy (red line).[2] Ignoring trading costs, a \$1 invested in the EOB strategy would have yielded \$4.74 which ranks the EOB strategy in the top 6% across the

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the EOB strategy would have yielded \$4.17 which ranks the EOB strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EOB relative to those. Panel A shows that the EOB strategy gross alphas fall between the 25 and 83 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EOB strategy has a positive net generalized alpha for five out of the five factor models. In these cases EOB ranks between the 52 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EOB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EOB with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EOB or at least to weaken the power EOB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EOB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EOB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EOB} EOB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EOB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EOB. Stocks are finally grouped into five EOB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EOB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EOB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EOB signal in these Fama-MacBeth regressions exceed 1.12, with the minimum t-statistic occurring when controlling for Predicted Analyst forecast error. Controlling for all six closely related anomalies, the t-statistic on EOB

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

is 1.27.

Similarly, Table 5 reports results from spanning tests that regress returns to the EOB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EOB strategy earns alphas that range from 25-39bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for Predicted Analyst forecast error. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EOB trading strategy achieves an alpha of 29bps/month with a t-statistic of 3.03.

7 Does EOB add relative to the whole zoo?

Finally, we can ask how much adding EOB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EOB signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes EOB grows to $3744.48.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EOB is available.

8 Conclusion

This study provides compelling evidence that Equity Operational Burden (EOB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EOB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of operational efficiency in equity valuation. The value-weighted long/short strategy based on EOB delivers an impressive annualized Sharpe ratio of 0.32 gross (0.31 net), indicating strong performance even after accounting for transaction costs. More importantly, the strategy's ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (36 bps/month gross, 29 bps/month net) suggests that EOB captures a distinct dimension of risk or mispricing not explained by conventional factors. The persistence of these results, with monthly alphas remaining statistically and economically significant even after controlling for six closely related strategies including volume-based and tail risk measures, further validates the unique information content embedded in the EOB signal.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The robust performance of EOB after transaction costs suggests it could be implementable in real-world portfolio construction, while its orthogonality to existing factors indicates potential diversification benefits when combined with other systematic strategies. The signal's economic intuition—that firms with higher operational burdens face greater challenges in generating shareholder value—provides a clear theoretical foundation that enhances its appeal to fundamental investors.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis accounts for transaction costs, implementation challenges such as market impact and capacity constraints in less liquid stocks require further investigation. Second, the stability of the EOB signal across different market regimes and its performance during periods of market stress deserve additional scrutiny. Third, the international applicability of our findings remains unexplored, as our analysis focuses exclusively on U.S. equities.

Future research should address these limitations while exploring several promising avenues. Investigating the underlying economic mechanisms driving the EOB premium through detailed fundamental analysis and earnings announcement studies could provide deeper insights into the signal's predictive power. Additionally, examining the interaction between EOB and corporate actions such as restructuring initiatives or operational improvements might reveal time-varying patterns in the signal's effectiveness. Finally, extending the analysis to international markets and exploring the signal's performance across different regulatory and accounting regimes would enhance our understanding of its global applicability and robustness.

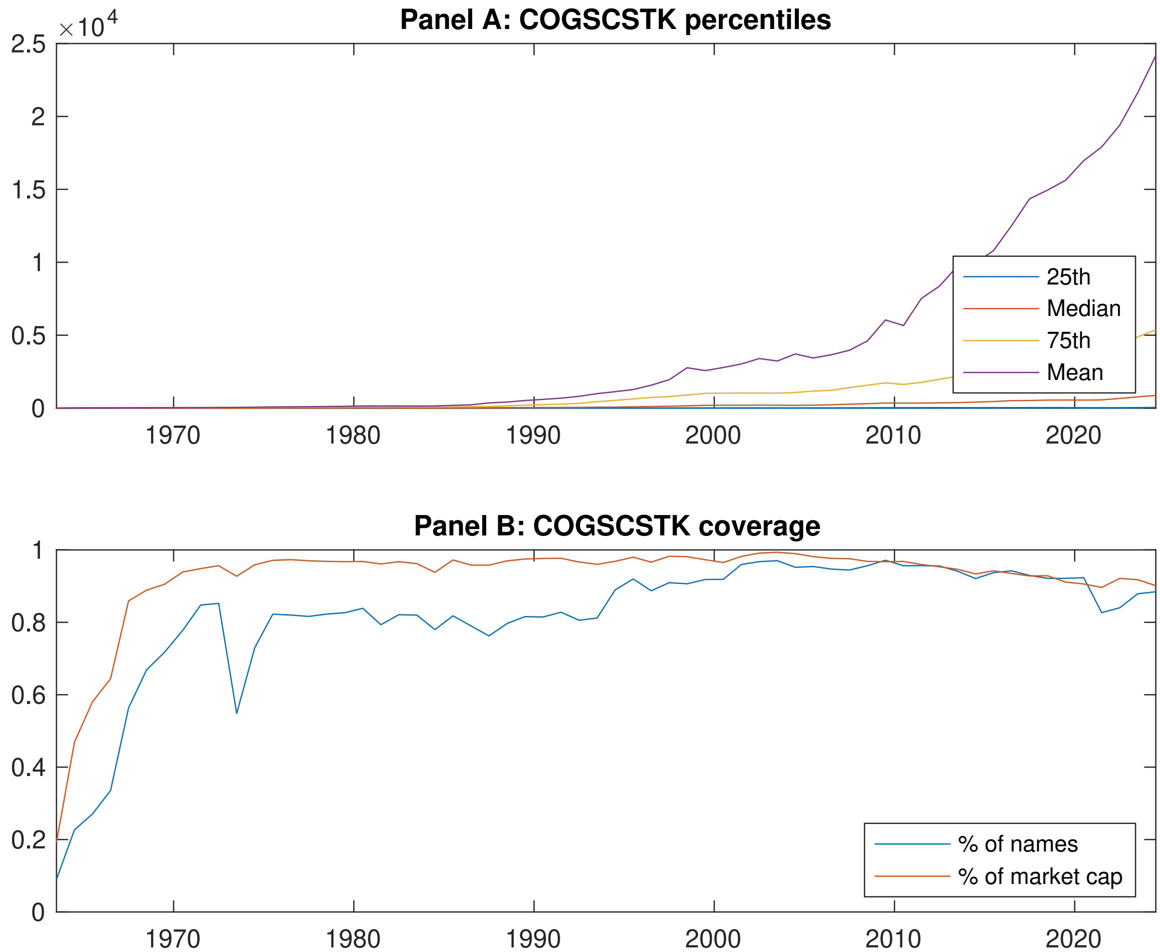


Figure 1: Times series of EOB percentiles and coverage.
This figure plots descriptive statistics for EOB. Panel A shows cross-sectional percentiles of EOB over the sample. Panel B plots the monthly coverage of EOB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EOB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EOB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48	0.57	0.52	0.62	0.77	0.29
	[3.31]	[3.51]	[3.05]	[3.16]	[3.91]	[2.52]
α_{CAPM}	0.00	0.01	-0.07	-0.06	0.10	0.10
	[0.06]	[0.32]	[-1.79]	[-1.12]	[1.70]	[0.96]
α_{FF3}	-0.06	-0.03	-0.06	-0.06	0.16	0.22
	[-1.36]	[-0.75]	[-1.51]	[-1.17]	[2.85]	[2.50]
α_{FF4}	-0.07	-0.06	-0.05	0.01	0.16	0.23
	[-1.50]	[-1.27]	[-1.17]	[0.28]	[2.87]	[2.59]
α_{FF5}	-0.15	-0.12	-0.06	0.02	0.21	0.36
	[-3.18]	[-2.86]	[-1.42]	[0.39]	[3.76]	[4.04]
α_{FF6}	-0.15	-0.13	-0.05	0.07	0.21	0.36
	[-3.14]	[-3.11]	[-1.13]	[1.51]	[3.67]	[3.97]
Panel B: Fama and French (2018) 6-factor model loadings for EOB-sorted portfolios						
β_{MKT}	0.89	1.00	1.01	1.09	1.06	0.18
	[77.92]	[99.01]	[100.49]	[94.06]	[78.37]	[8.16]
β_{SMB}	-0.11	-0.01	-0.01	0.05	0.13	0.24
	[-6.96]	[-0.88]	[-0.91]	[2.79]	[6.62]	[7.84]
β_{HML}	0.16	0.07	-0.04	0.05	-0.11	-0.27
	[7.27]	[3.61]	[-2.17]	[2.09]	[-4.23]	[-6.50]
β_{RMW}	0.16	0.16	-0.01	-0.09	-0.05	-0.22
	[7.31]	[7.84]	[-0.63]	[-3.98]	[-1.99]	[-5.11]
β_{CMA}	0.14	0.16	0.02	-0.19	-0.17	-0.31
	[4.37]	[5.72]	[0.73]	[-5.70]	[-4.51]	[-5.14]
β_{UMD}	0.00	0.02	-0.02	-0.08	0.00	0.00
	[0.01]	[1.73]	[-1.62]	[-6.70]	[0.17]	[0.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	860	629	720	734	783	
me ($\$10^6$)	2287	1948	1705	1576	2319	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the EOB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29	0.10	0.22	0.23	0.36	0.36
			[2.52]	[0.96]	[2.50]	[2.59]	[4.04]	[3.97]
Quintile	NYSE	EW	0.22	0.05	0.04	0.11	0.02	0.08
			[2.48]	[0.62]	[0.57]	[1.65]	[0.27]	[1.22]
Quintile	Name	VW	0.27	0.06	0.18	0.21	0.31	0.32
			[2.21]	[0.53]	[2.03]	[2.37]	[3.41]	[3.55]
Quintile	Cap	VW	0.23	0.03	0.17	0.21	0.30	0.33
			[1.92]	[0.27]	[1.82]	[2.25]	[3.22]	[3.45]
Decile	NYSE	VW	0.38	0.14	0.33	0.32	0.44	0.42
			[2.59]	[1.06]	[3.08]	[2.92]	[4.04]	[3.80]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.28	0.08	0.19	0.20	0.29	0.29
			[2.40]	[0.78]	[2.16]	[2.23]	[3.30]	[3.27]
Quintile	NYSE	EW	0.12					
			[1.28]					
Quintile	Name	VW	0.25	0.04	0.15	0.17	0.23	0.24
			[2.08]	[0.33]	[1.67]	[1.88]	[2.61]	[2.71]
Quintile	Cap	VW	0.22	0.01	0.14	0.17	0.23	0.25
			[1.81]	[0.14]	[1.53]	[1.79]	[2.51]	[2.67]
Decile	NYSE	VW	0.36	0.12	0.29	0.29	0.36	0.35
			[2.47]	[0.90]	[2.69]	[2.64]	[3.32]	[3.22]

Table 3: Conditional sort on size and EOB

This table presents results for conditional double sorts on size and EOB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EOB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EOB and short stocks with low EOB .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	EOB Quintiles					EOB Strategies					
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.48	0.71	0.82	0.74	0.93	0.44	0.29	0.25	0.31	0.19	0.25
	[2.18]	[3.10]	[3.27]	[2.97]	[3.41]	[3.70]	[2.60]	[2.23]	[2.73]	[1.64]	[2.16]
(2)	0.63	0.80	0.83	0.78	0.84	0.21	0.06	0.12	0.10	0.09	0.07
	[3.20]	[3.73]	[3.48]	[3.23]	[3.45]	[1.96]	[0.58]	[1.29]	[1.05]	[0.93]	[0.77]
(3)	0.53	0.68	0.82	0.81	0.84	0.31	0.11	0.18	0.16	0.15	0.14
	[3.12]	[3.38]	[3.85]	[3.61]	[3.62]	[2.56]	[0.98]	[1.94]	[1.72]	[1.59]	[1.44]
(4)	0.57	0.61	0.69	0.73	0.80	0.23	0.04	0.16	0.15	0.17	0.16
	[3.48]	[3.33]	[3.43]	[3.56]	[3.73]	[2.02]	[0.44]	[1.87]	[1.68]	[1.98]	[1.80]
(5)	0.48	0.48	0.47	0.56	0.69	0.21	0.01	0.15	0.20	0.32	0.35
	[3.35]	[3.01]	[2.81]	[3.08]	[3.62]	[1.61]	[0.07]	[1.42]	[1.87]	[3.02]	[3.25]

Panel B: Portfolio average number of firms and market capitalization

Size quintiles	EOB Quintiles: Average n					EOB Quintiles: Average market capitalization ($\$10^6$)				
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	418	420	419	418	417	32	33	33	36	40
(2)	116	116	116	115	116	59	59	59	60	59
(3)	83	82	82	82	82	102	101	101	101	101
(4)	68	68	68	68	68	216	214	212	212	206
(5)	61	61	61	61	61	1564	1368	1405	1383	2078

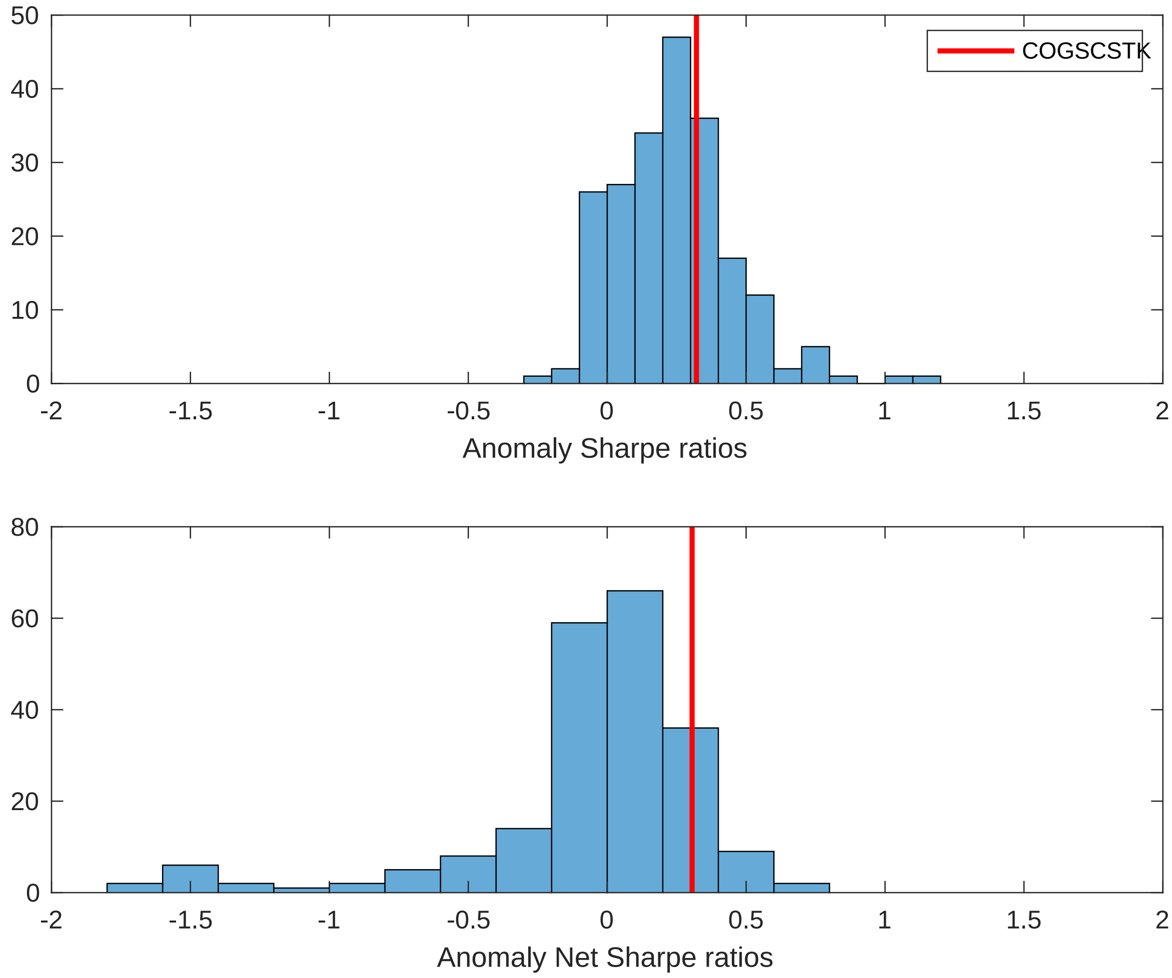


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EOB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

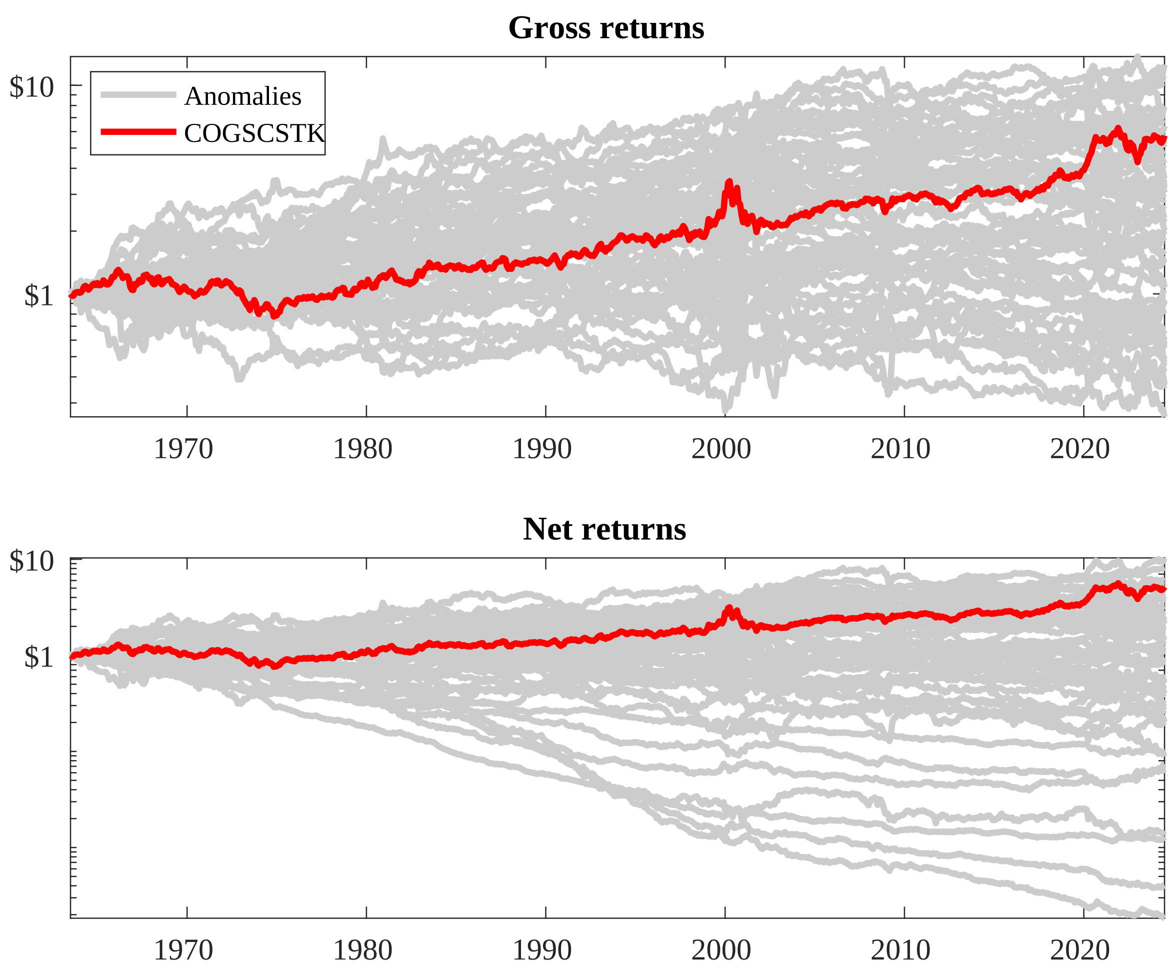
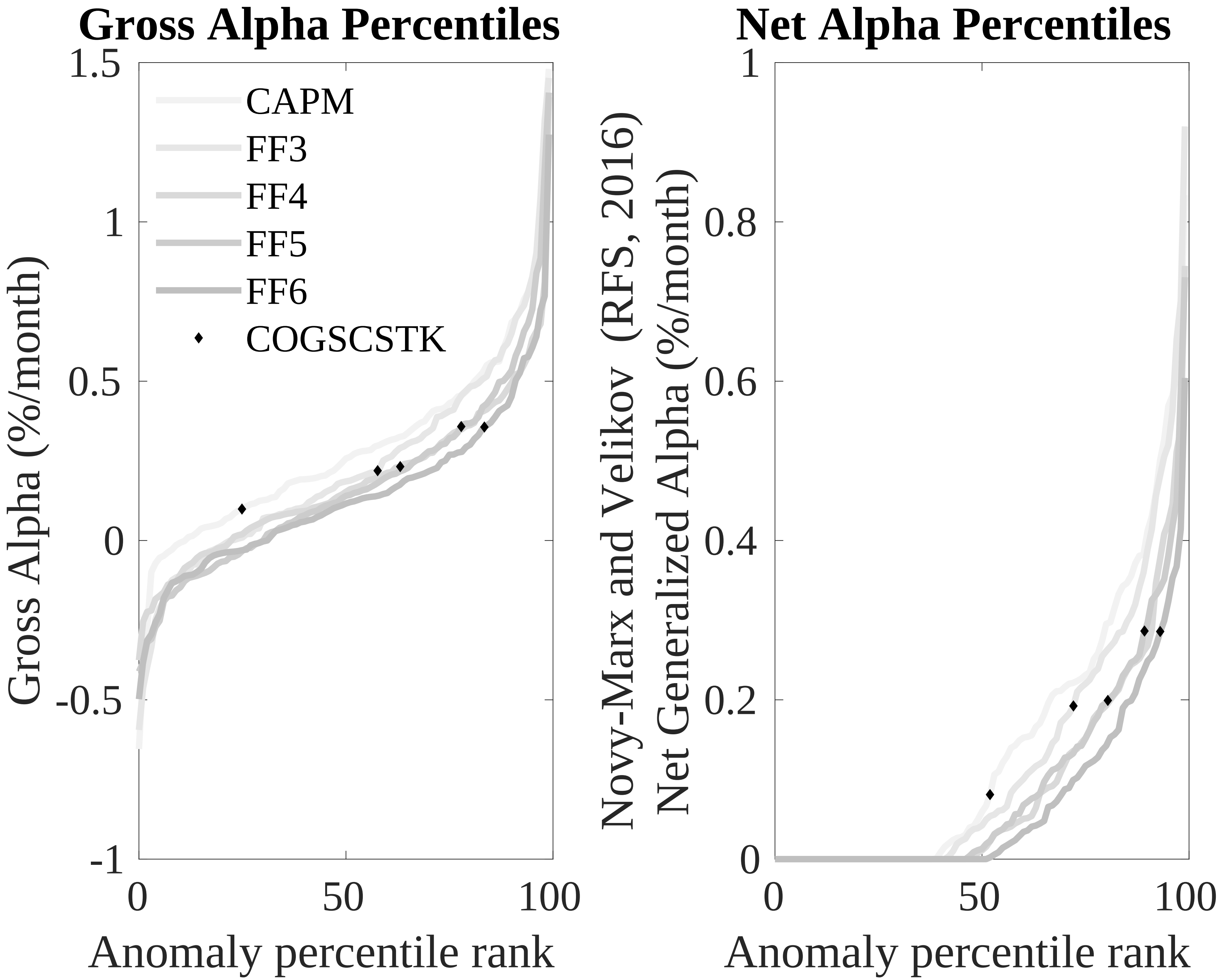
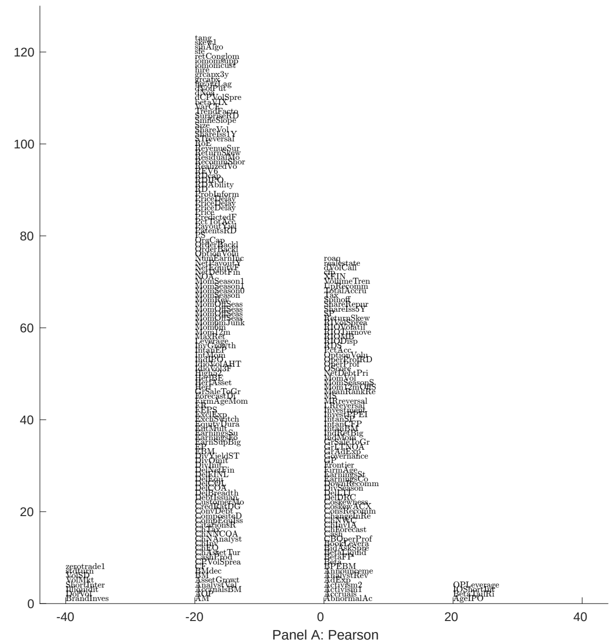


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EOB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
COGSCSTK
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



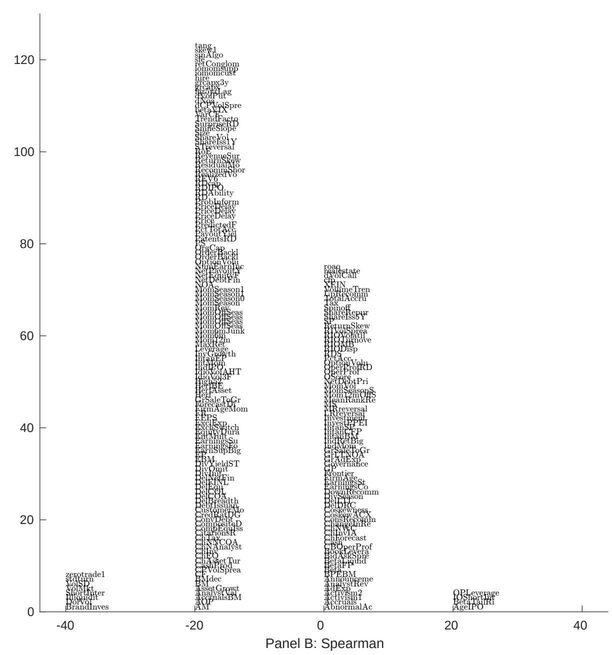


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EOB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

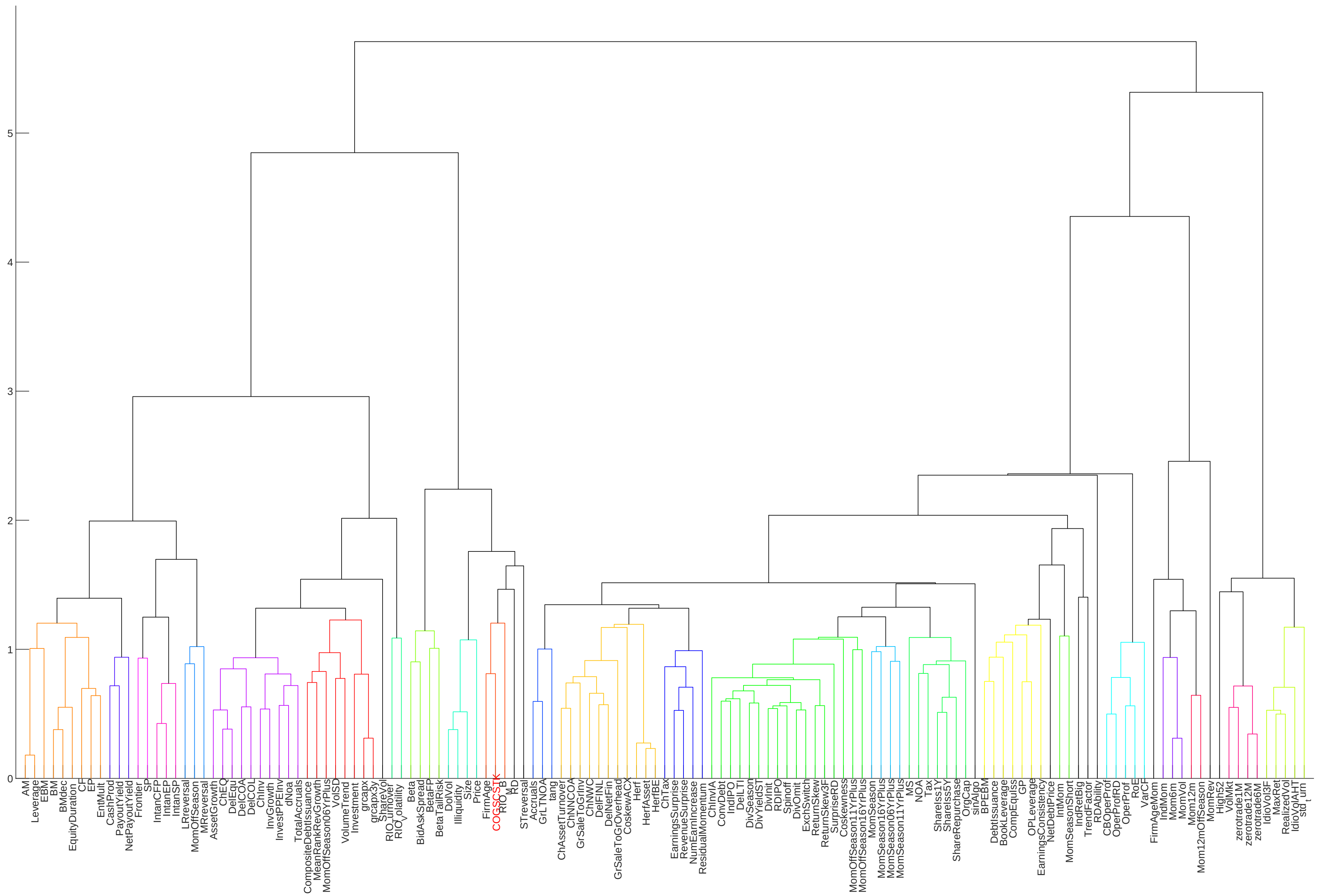


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

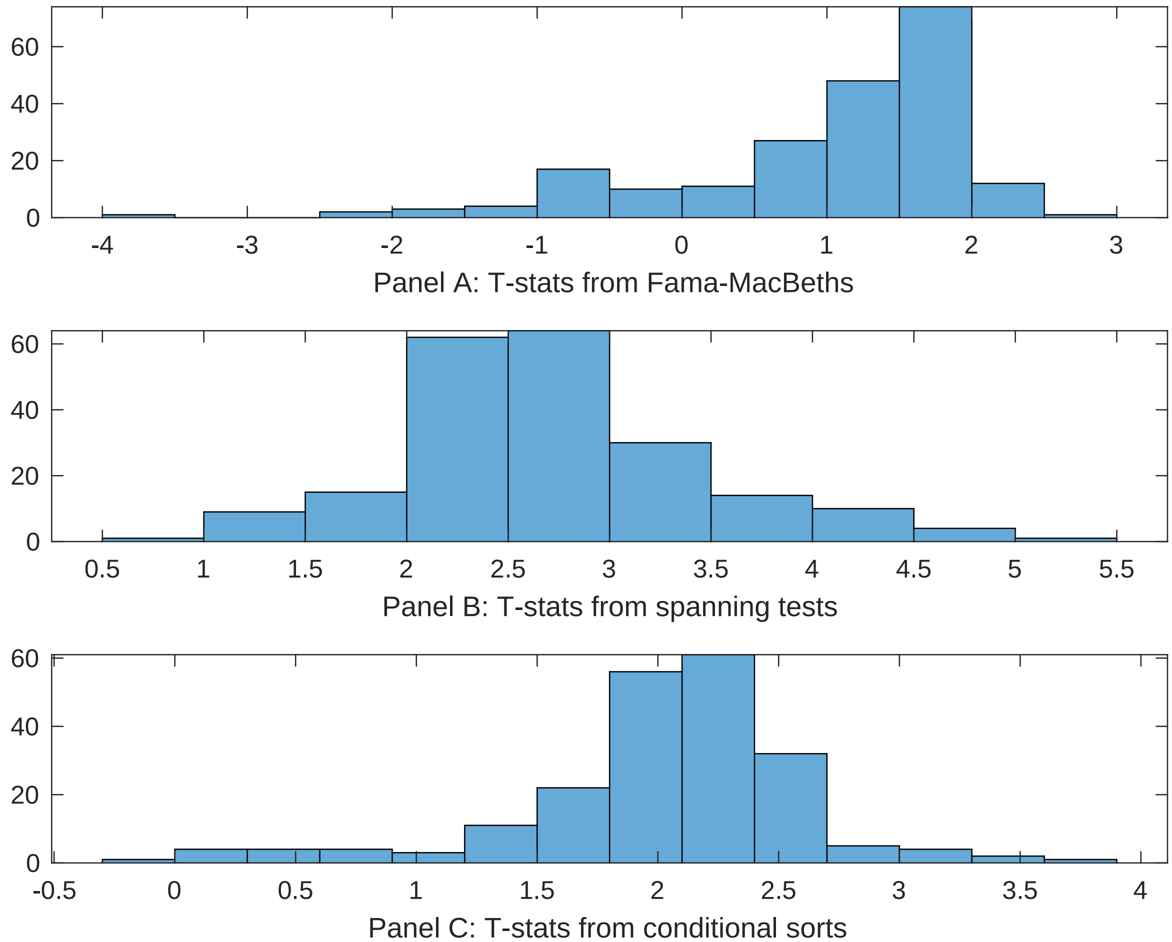


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EOB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EOB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EOB} EOB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EOB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EOB. Stocks are finally grouped into five EOB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EOB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EOB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EOB} EOB_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Tail risk beta, Predicted Analyst forecast error. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.11	0.10	0.10	0.97	0.12	0.11
	[7.36]	[4.76]	[4.61]	[4.64]	[6.05]	[5.33]	[6.28]
EOB	0.60	0.85	0.85	0.87	0.42	0.28	0.30
	[1.40]	[1.74]	[1.76]	[1.79]	[1.17]	[1.12]	[1.27]
Anomaly 1	0.20						0.15
	[1.53]						[2.47]
Anomaly 2		0.23					0.61
		[1.38]					[0.26]
Anomaly 3			0.19				-0.65
			[2.87]				[-0.72]
Anomaly 4				0.98			0.39
				[2.98]			[0.23]
Anomaly 5					0.51		0.31
					[3.13]		[1.70]
Anomaly 6						0.17	0.11
						[0.65]	[0.58]
# months	727	727	727	727	727	492	487
$\bar{R}^2$(%)	2	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the EOB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EOB} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Tail risk beta, Predicted Analyst forecast error. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.27	0.32	0.36	0.33	0.25	0.39	0.29
	[3.16]	[3.70]	[4.15]	[3.83]	[2.88]	[3.87]	[3.03]
Anomaly 1	-27.60						-13.40
	[-10.54]						[-1.97]
Anomaly 2		-23.99					0.29
		[-9.25]					[0.04]
Anomaly 3			-22.17				-1.67
			[-8.46]				[-0.17]
Anomaly 4				-23.15			-9.70
				[-8.80]			[-1.15]
Anomaly 5					22.89		6.44
					[9.12]		[1.98]
Anomaly 6						-24.06	-15.02
						[-7.94]	[-4.81]
mkt	5.15	7.56	8.19	8.05	8.26	13.44	0.63
	[2.20]	[3.26]	[3.49]	[3.46]	[3.61]	[5.49]	[0.23]
smb	8.22	15.80	21.14	23.73	15.50	23.63	11.71
	[2.48]	[5.07]	[7.03]	[8.01]	[4.95]	[6.47]	[2.79]
hml	-18.82	-20.35	-20.53	-21.00	-28.66	-14.95	-14.86
	[-4.80]	[-5.12]	[-5.12]	[-5.27]	[-7.25]	[-3.15]	[-3.26]
rmw	-6.79	-6.68	-7.31	-7.97	-13.13	-9.87	3.77
	[-1.63]	[-1.56]	[-1.68]	[-1.86]	[-3.22]	[-2.07]	[0.78]
cma	-16.48	-18.72	-18.90	-17.58	-15.35	-44.38	-28.06
	[-2.81]	[-3.15]	[-3.14]	[-2.92]	[-2.53]	[-6.80]	[-4.31]
umd	6.32	-0.05	0.63	1.26	3.46	1.37	8.28
	[3.07]	[-0.03]	[0.31]	[0.62]	[1.70]	[0.60]	[3.14]
# months	728	728	728	728	728	492	488
$\bar{R}^2$(%)	54	52	51	52	52	61	66

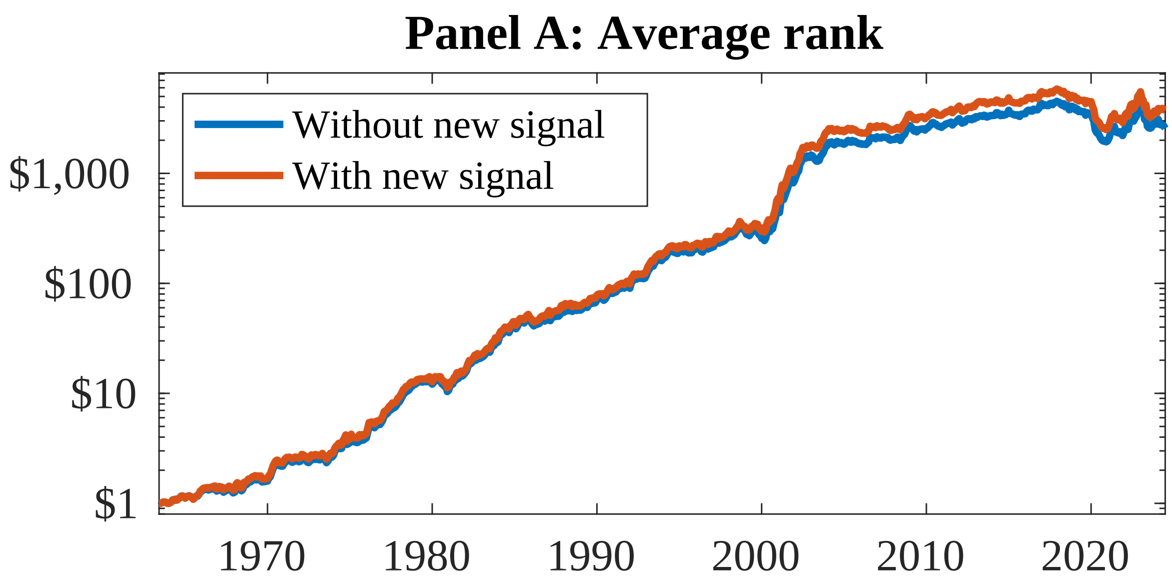


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EOB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.

Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.

Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Jagannathan, R. and Wang, Z. (1996). The conditional capm and the cross-section of expected returns. *Journal of Finance*, 51(1):3–53.

Lettau, M. and Ludvigson, S. (2001a). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.

Lettau, M. and Ludvigson, S. (2001b). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Parker, J. A. and Julliard, C. (2005). Consumption risk and the cross section of expected returns. *Journal of Political Economy*, 113(1):185–222.

Santos, T. and Veronesi, P. (2010). Habit formation, the cross section of stock returns and the cash-flow risk puzzle. *Journal of Financial Economics*, 98(2):385–413.

Wachter, J. A. (2013). Can time-varying risk of rare disasters explain aggregate stock market volatility? *Journal of Finance*, 68(3):987–1035.

Yogo, M. (2006). A consumption-based explanation of expected stock returns. *Journal of Finance*, 61(2):539–580.

Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.